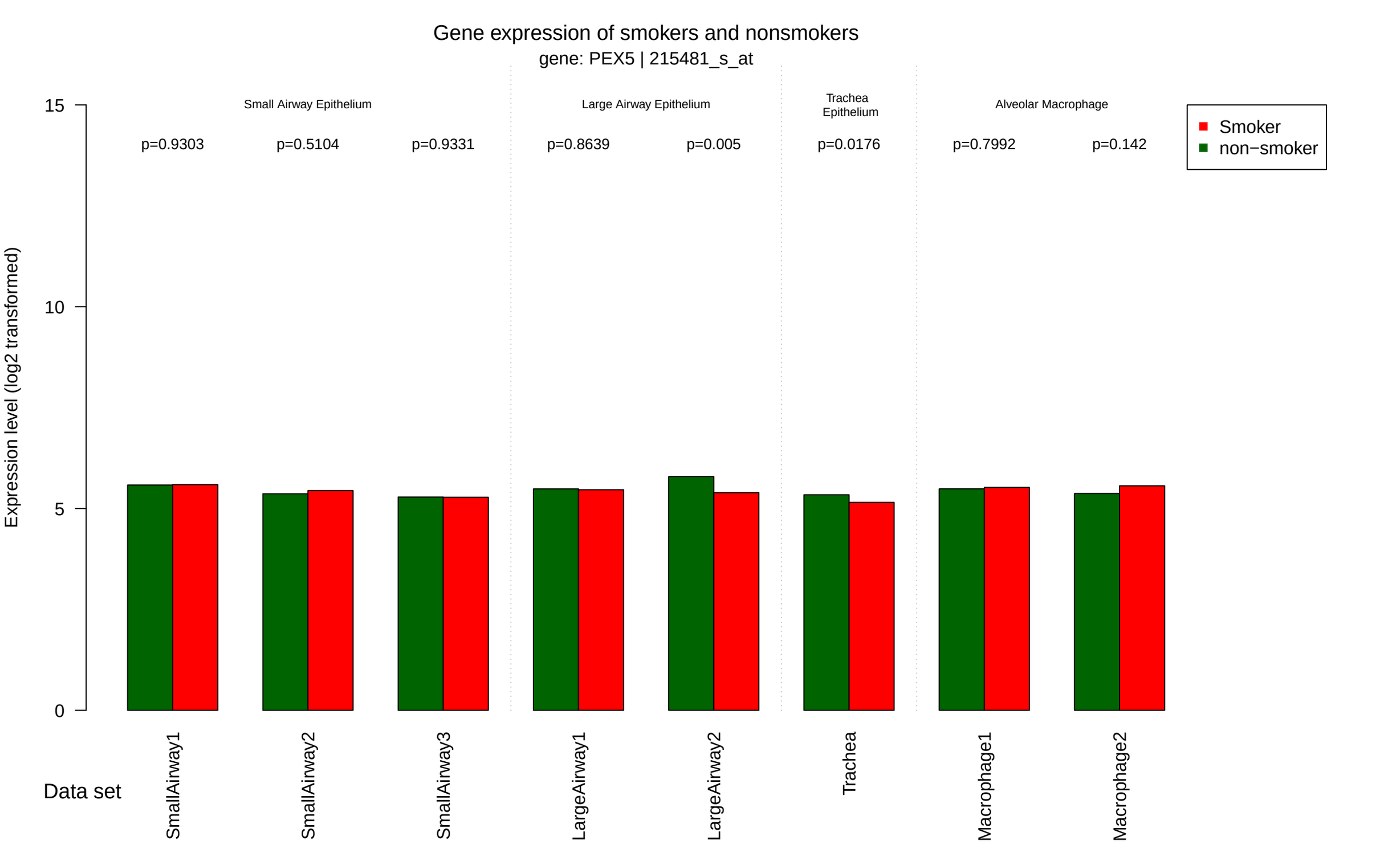
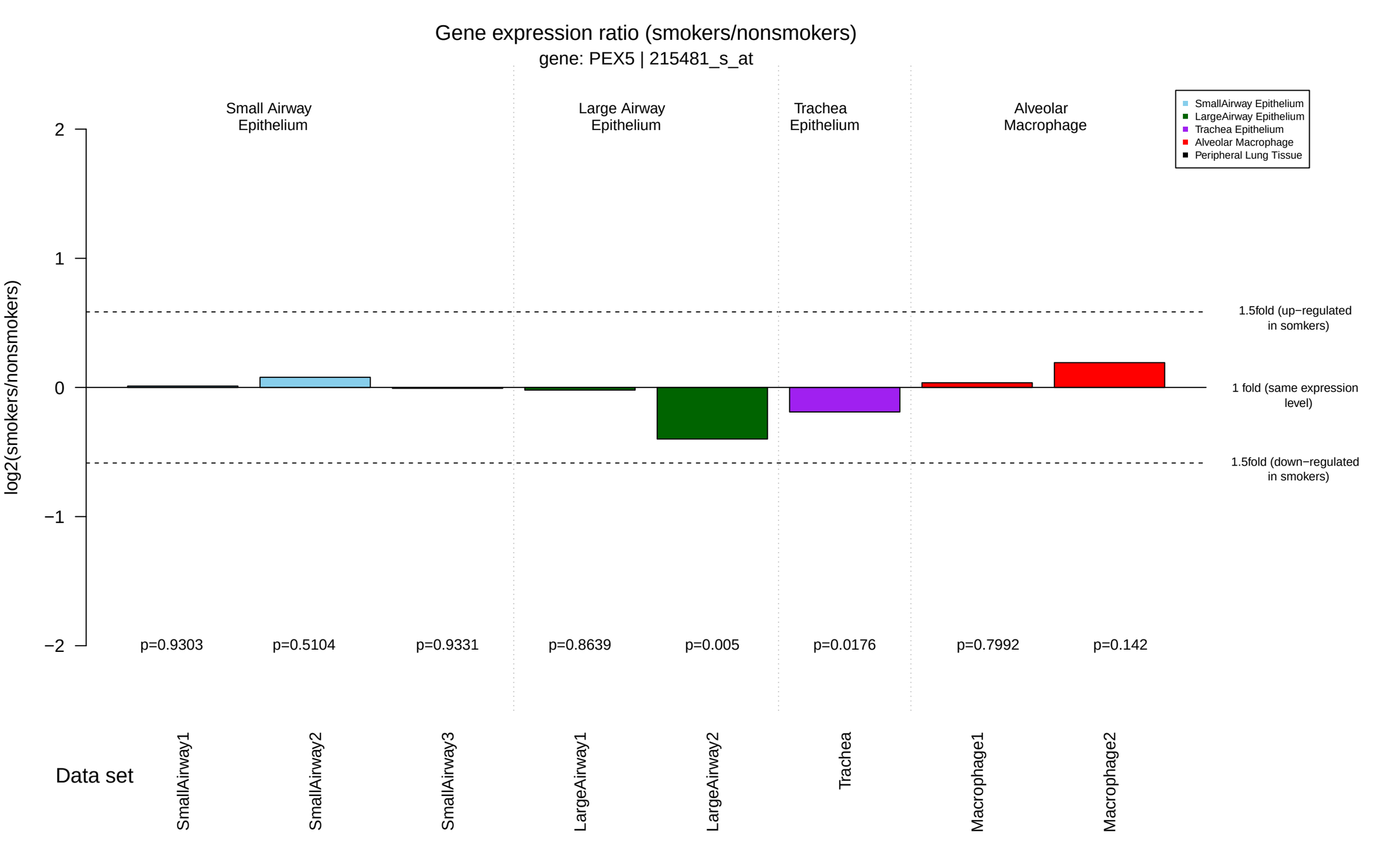
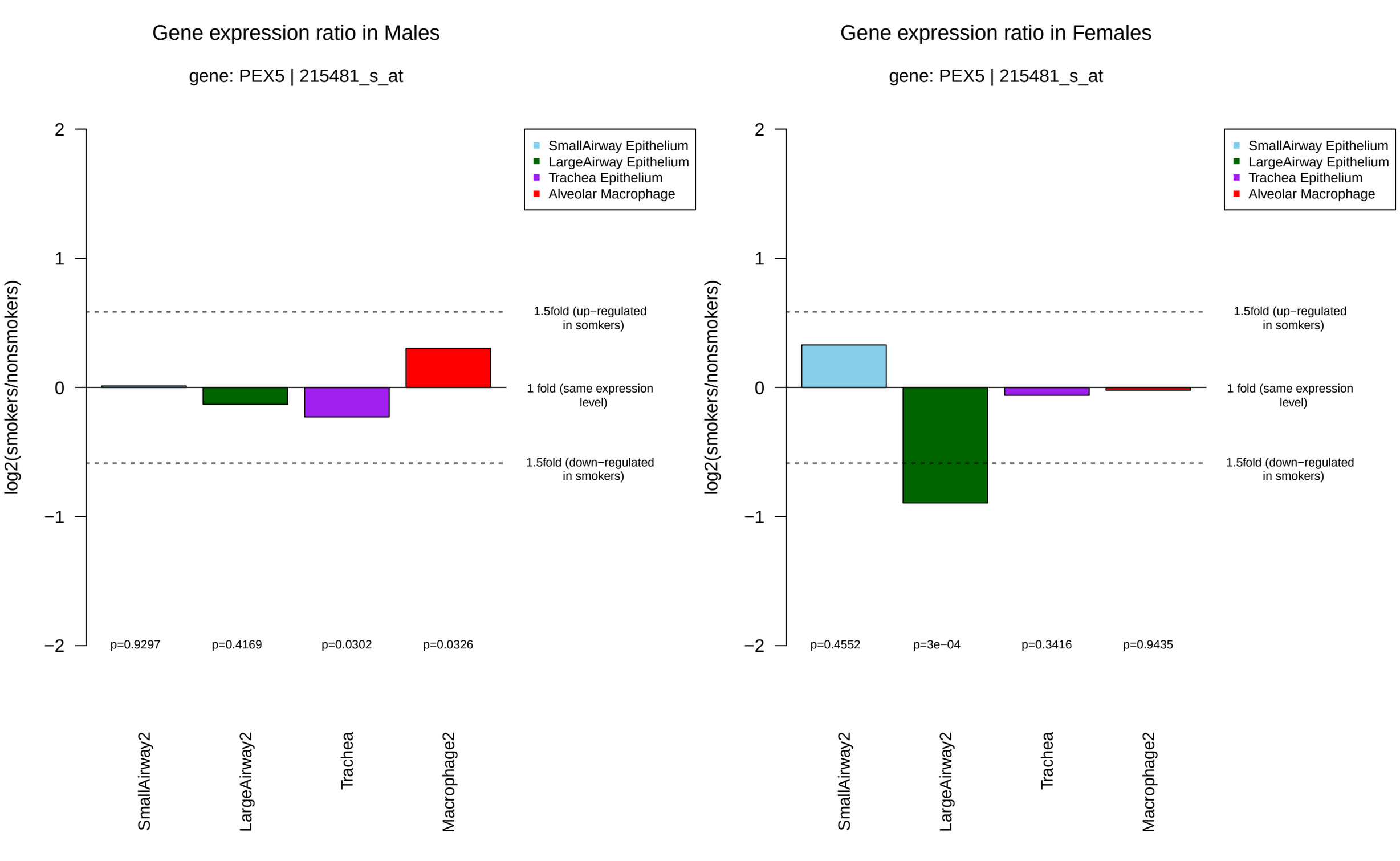
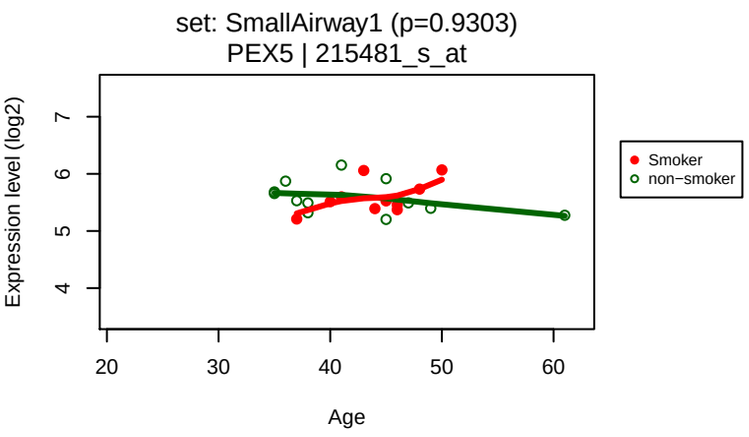
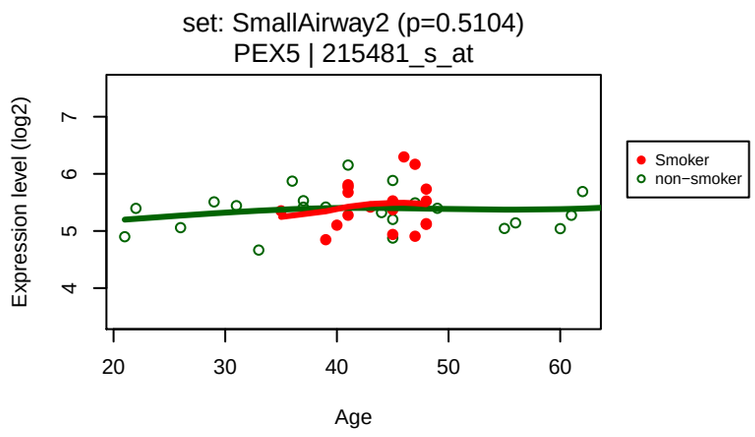
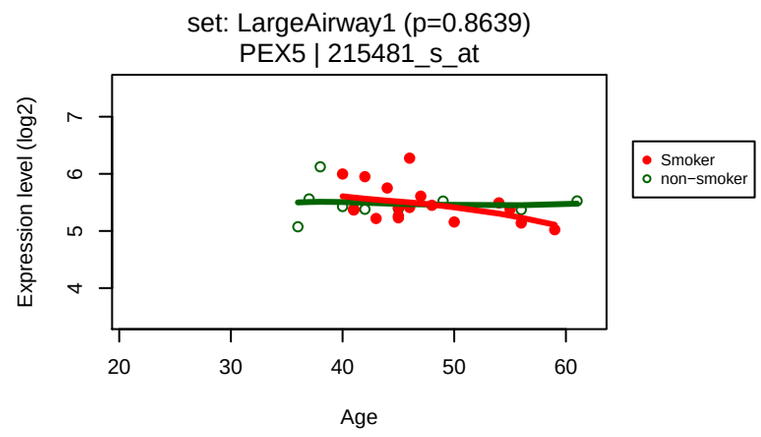

Gene expression of smokers and nonsmokers
gene: PEX5 | 215481_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9303
p=0.5104
p=0.9331
p=0.8639
p=0.005
p=0.0176
p=0.7992
p=0.142
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

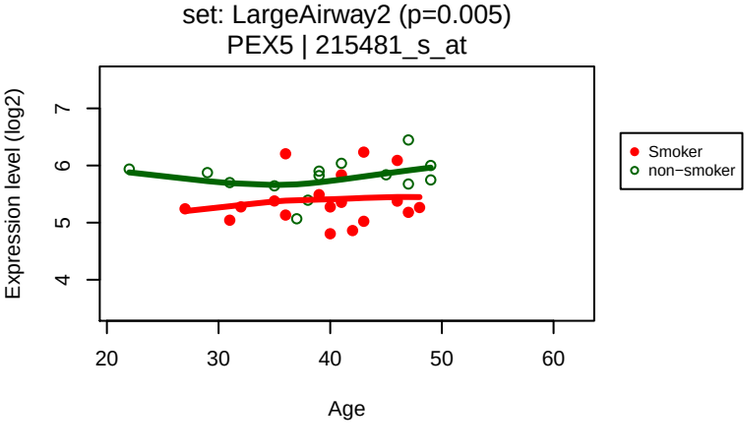

Gene expression ratio (smokers/nonsmokers)
gene: PEX5 | 215481_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9303
p=0.5104
p=0.9331
p=0.8639
p=0.005
p=0.0176
p=0.7992
p=0.142
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

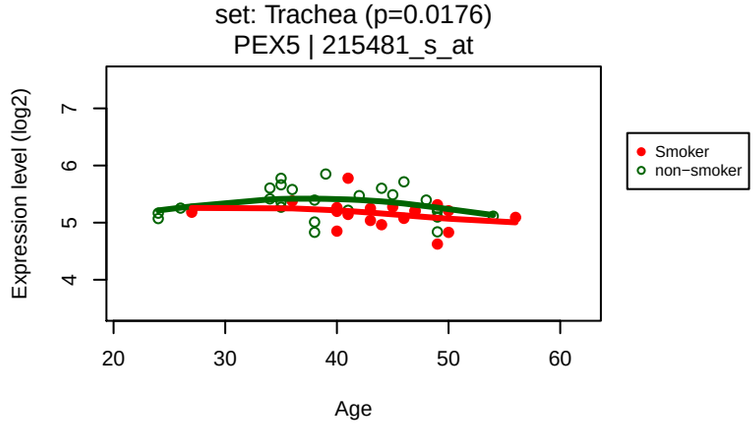

Gene expression ratio in Males
gene: PEX5 | 215481_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9297
p=0.4169
p=0.0302
p=0.0326
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PEX5 | 215481_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4552
p=3e−04
p=0.3416
p=0.9435
SmallAirway2
LargeAirway2
Trachea
Macrophage2

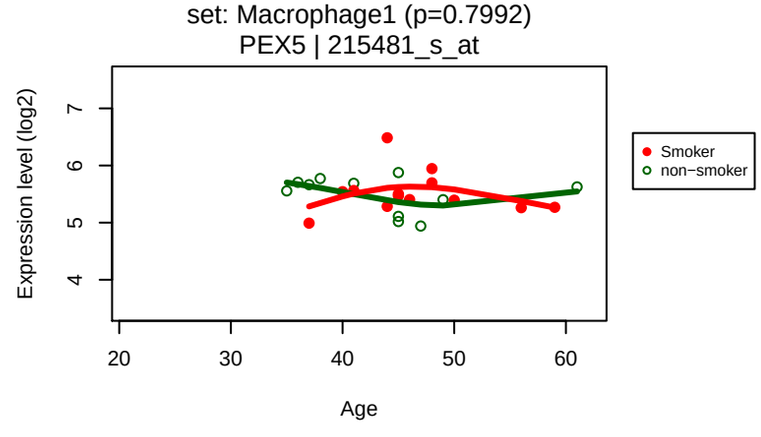

set: SmallAirway1 (p=0.9303)
PEX5 | 215481_s_at
Expression level (log2)
Age
Smoker
non−smoker

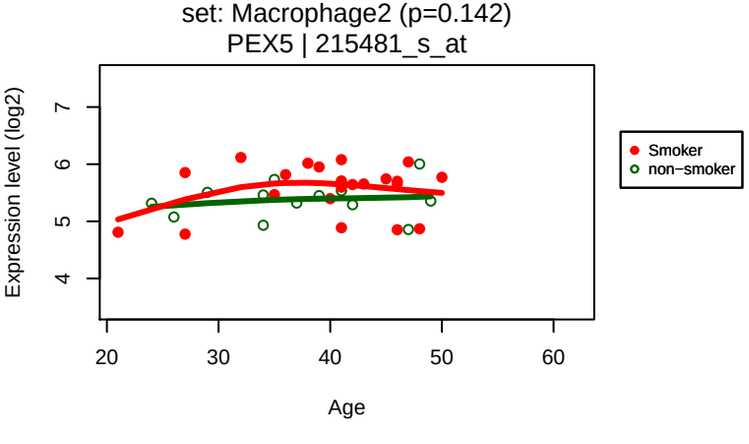

set: SmallAirway2 (p=0.5104)
PEX5 | 215481_s_at
Expression level (log2)
Age
Smoker
non−smoker

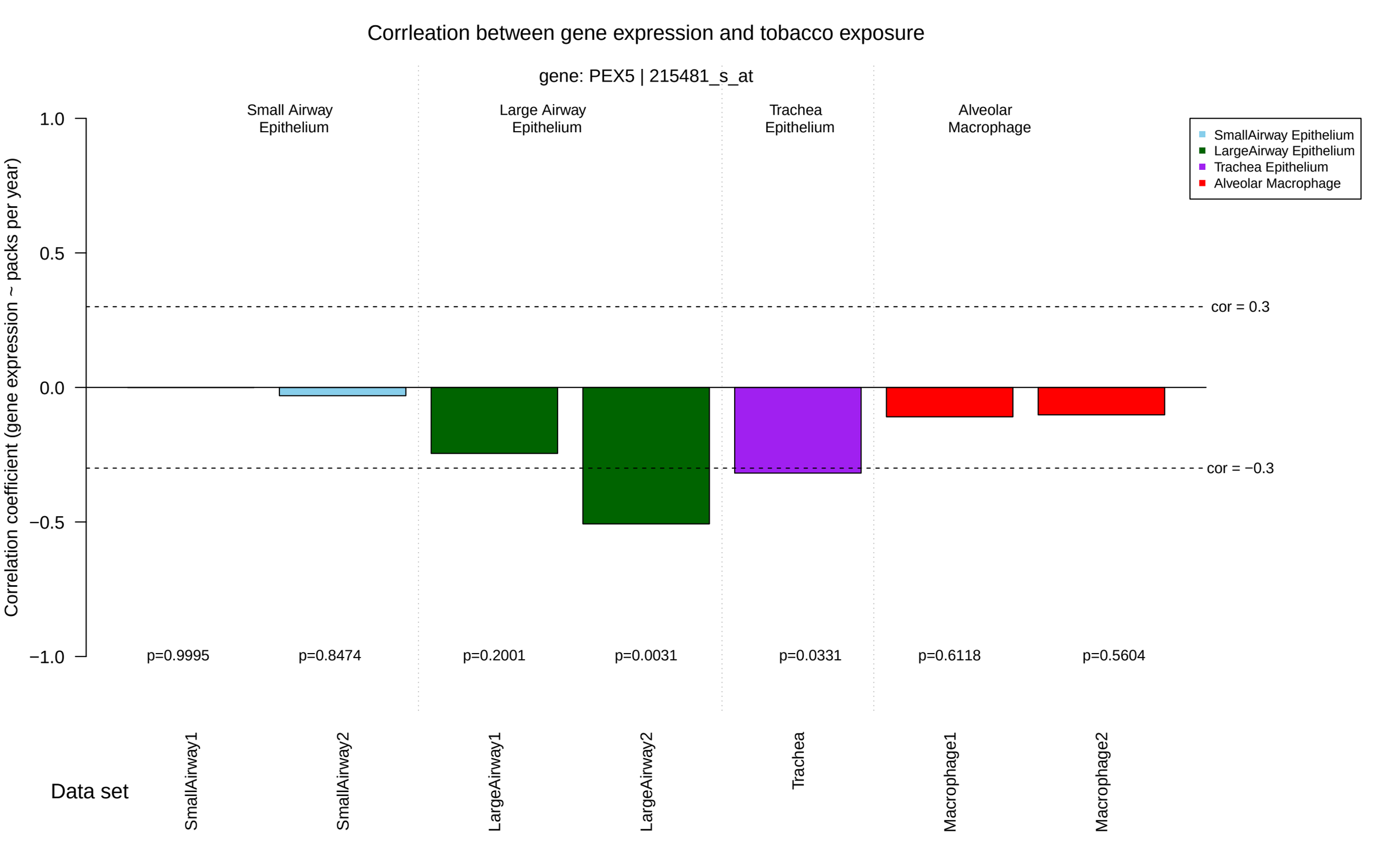

set: LargeAirway1 (p=0.8639)
PEX5 | 215481_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.005)
PEX5 | 215481_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0176)
PEX5 | 215481_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7992)
PEX5 | 215481_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.142)
PEX5 | 215481_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PEX5 | 215481_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9995
p=0.8474
p=0.2001
p=0.0031
p=0.0331
p=0.6118
p=0.5604
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2